

Incident Runbook Leave Management System

Severity	SEV 2 by default • Raise to SEV 1 for security, data-integrity, or broad service impact
Critical flow	Submit leave request → approval workflow → balance update → employee notification

Alarm meaning Error-rate alarm: more than 5% Lambda errors for two consecutive 5-minute periods. Latency alarm: Lambda P95 duration above 3,000 ms for two consecutive 5-minute periods.

POSSIBLE CAUSES

- 401: missing, invalid, or expired Cognito JWT; approval HMAC token expired
- 403: valid user lacks the required Manager/HR group or route permission
- DynamoDB AccessDenied on the single-writer balance update, conditional conflict, or throttling
- Step Functions state failure, timeout, or malformed state input
- SES/SNS failure, cold start, workflow wait/retry, or downstream latency

BUSINESS IMPACT

Employees may be unable to submit requests; approvals or leave balances may be delayed or inconsistent; notifications may not reach managers or employees.

FIRST RESPONSE

1. Acknowledge the SNS alert; note alarm name, UTC time, region, and affected function.
2. For 401, validate Bearer header, 60-minute JWT expiry/issuer/audience; issue a fresh approval token instead of reviving an expired one. For 403, verify Manager/HR group claims and authorizer policy.
3. Search logs using the correlation ID; inspect the X-Ray trace and the related Step Functions execution.
4. For DynamoDB AccessDenied, identify the denied action and verify the execution role has the exact table/index ARN; do not broaden access to Resource *.
5. Stop unsafe retries if a balance could be applied twice. Correct the issue, then retry only the failed workflow using its idempotency key.

RECOVERY VERIFICATION

- Complete one safe test: submit → approve → balance update → notification.
- Confirm alarm returns to OK for two periods; no new errors or throttles appear.
- Verify the request status and leave balance match, with only one notification sent.

ESCALATION CONDITIONS

- Error/latency remains breached for 15 minutes after first response.
- Multiple users are affected, data integrity is uncertain, or duplicate balance updates occurred.
- IAM/KMS access, AWS regional service health, or security exposure is suspected.

Recovery evidence to attach Alarm history; Log Insights result; X-Ray trace ID; Step Functions execution ARN; test request ID; incident timeline.